

Description

At 9:00 AM a nurse sees the patient and performs a vital sign assessment, noting that the child currently has a temperature of 101.2 with a productive cough and right ear inflammation and fluid build-up.

At 9:10 AM, the physician suspects the patient has influenza with other respiratory manifestations and assigns the patient a working diagnoses of influenza with other respiratory manifestations (ICD-10 CM diagnosis code of Influenza due to unidentified influenza virus with other respiratory manifestations J11.1), and Acquired stenosis of external ear canal secondary to inflammation and infection, unspecified ear (ICD-10 CM diagnosis code of H61.329) and documents the information in the patient's electronic medical record. At 9:15 AM the physician orders a rapid influenza test, chest x-ray, and a treatment. The patient's vital signs gradually return to normal. A nurse weighs the patient, measures his length, and charts the data in the patient's record. This information is used by the physician when ordering the take-home prescription.

NIST Hospital's Urgent Care is an outpatient facility operated by NIST Hospital that routinely sends electronic syndromic surveillance data to the the City Health Department (NIST PH SS FAC 1) in accordance with a city regulation. At 9:15 AM, NIST Hospital's electronic health record module for syndromic surveillance data assembles and transmits an Update ADT message with the working diagnoses and the height and weight information for this patient encounter on behalf of the Urgent Care facility as part of an automated process that transmits new or updated content every 15 minutes. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity.

Comments

Dates and times are provided in this test scenario to illustrate the sequence of clinical and messaging events. Since the exact dates and times are not reproducible when modeling the test scenario with HIT, only date and time format will be validated within tester submitted test data

Pre-condition

A04-Registration message is sent before the A08-Update message.

Post-Condition

Patient record is updated with procedure results, diagnoses and patient height/weight. An ADT*A08 message has been generated with this updated content.

Test Objectives

This test step examines a Health IT Module's ability to create ADT^A08 Update message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements.

The following test objectives and variants are supported by this test case:

- Setting is Urgent Care.
- Chief complaint is submitted as free-text.
- **Patient name is de-identified in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in a de-identified manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Pediatric case where age is measured in months.
- Updates to Diagnoses
- Multiple Races
- Insurance is known

Evaluation Criteria

Message Validation Report

Diagnosis code (DG1-3) validation is validated as follows:

The table below summarizes the format representations that are accepted for ONC Health IT Module Certification. Rows 1 and 2 show the formats as required by the implementation guide and standard terminology rules and best practices. The last 3 columns show acceptable alternative formats. For this test case one of the following values shall be present in the message:

DG1(1)-3

Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
J111	Influenza due to unidentified influenza virus with other respiratory manifestations	I10C	J111^Influenza due to unidentified influenza virus with other respiratory manifestations^I10C	Yes
442438000	Influenza caused by Influenza A virus (disorder)	SCT	442438000^Influenza caused by Influenza A virus (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
J11.1	Influenza due to unidentified influenza virus with other respiratory manifestations	I10C	J11.1^Influenza due to unidentified influenza virus with other respiratory manifestations^I10C	Yes
J111	Influenza due to unidentified influenza virus with other respiratory manifestations	I10	J111^Influenza due to unidentified influenza virus with other respiratory manifestations^I10	No
J11.1	Influenza due to unidentified influenza virus with other respiratory manifestations	I10	J11.1^Influenza due to unidentified influenza virus with other respiratory manifestations^I10	No

328531000119104	Upper respiratory tract infection caused by Influenza A (disorder)	SCT	328531000119104^Upper respiratory tract infection caused by Influenza A (disorder)^SCT	Yes
DG1(2)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
H61329	Acquired stenosis of external ear canal secondary to inflammation and infection, unspecified ear	I10C	H61329^Acquired stenosis of external ear canal secondary to inflammation and infection, unspecified ear^I10C	Yes
1090241000119103	Acquired stenosis of external ear canal secondary to infection (disorder)	SCT	1090241000119103^Acquired stenosis of external ear canal secondary to infection (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
H61.329	Acquired stenosis of external ear canal secondary to inflammation and infection, unspecified ear	I10C	H61.329^Acquired stenosis of external ear canal secondary to inflammation and infection, unspecified ear^I10C	Yes
H61329	Acquired stenosis of external ear canal secondary to inflammation and infection, unspecified ear	I10	H61329^Acquired stenosis of external ear canal secondary to inflammation and infection, unspecified ear^I10	No
H61.329	Acquired stenosis of external ear canal secondary to inflammation and infection, unspecified ear	I10	H61.329^Acquired stenosis of external ear canal secondary to inflammation and infection, unspecified ear ^I10	No
55719006	Acquired stenosis of external ear canal secondary to inflammation (disorder)	SCT	55719006^Acquired stenosis of external ear canal secondary to inflammation (disorder)^SCT	Yes
55989009	Acquired stenosis of external ear canal (disorder)	SCT	55989009^Acquired stenosis of external ear canal (disorder)^SCT	Yes

The NIST column indicates what the NIST Tool Validation will accept in context-based testing (used for ONC Certification). A 'Yes' in the NIST column signifies that the tool will accept the format and not throw an error. A 'No' in the NIST column signifies that the tool will throw an error (specifically for DG1-3.3). However, based on the recommendation from the CDC SMEs and approval from ASTP/ONC the format is acceptable. The NIST test cases documents the instances where the test proctor can grant an exception when the tool issues an error.

Notes

ICD-10 CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case. If the HIT developer uses different but equivalent ICD-10 CM codes than the ones provided, the Tester may ignore errors generated by the Test Tool related to incorrect code when the ICD-10 CM code data used in the message are determined via visual inspection to be valid data.

ICD-10 CM codes are acceptable with or without decimals.

Although the other units of measure for patient age are acceptable in general (and the Context-free validation accepts any of the valid units of measure for age), this Scenario specifies that "mo" for months be used in this message for the Context-based validation. Messages in this test case will be used in a Manual Test Step under functional requirement testing.